Adaptive Keyframe Sampling for Mixed Video Understanding

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Abstract

In the realm of video content moderation, Multimodal Large Language Models (MLLMs) face challenges processing long videos due to limited context windows. This project proposes a novel plug-and-play keyframe selection module. Inspired by prior works on AI-based moderation and keyframe sampling, the module formulates keyframe selection as an optimization problem, balancing relevance to textual prompts and temporal coverage of video content.It employs recursive divide-and-conquer and greedy strategies, reducing the time complexity from $\mathcal{O}(n \log n)$ to $\mathcal{O}(n)$ in the best condition through partial sorting. Experiments on LongVideoBench and a self-constructed mixed video dataset show that the proposed approach outperforms traditional sampling methods. Additionally, a revised hierarchical top-k segmentation algorithm improves the module's robustness and interpretability.The study effectively integrates course concepts like recursion, greedy algorithms, and complexity analysis into practical video processing, advancing the efficiency of video content moderation systems.

1. Introduction

In the era of exponentially growing user-generated video content, efficient and accurate video content moderation has become a critical challenge. Automated systems leveraging artificial intelligence are essential for filtering inappropriate or harmful content, but their effectiveness hinges on the ability to process video data efficiently. A key bottleneck in this pipeline is the limited context capacity of Multimodal Large Language Models (MLLMs) when handling long videos: for example, long-form videos exceeding one hour can contain tens of thousands of frames, while MLLMs like LLaVA-Video typically support only hundreds of frames in their context windows [11], necessitating intelligent keyframe selection to balance information richness and computational feasibility.

This project addresses the problem of optimizing keyframe selection for MLLMs in long video understanding tasks. Inspired by prior works on AI-based video moderation (e.g., SafeWatch's policy-aware frame pruning [1]) and adaptive keyframe sampling (e.g., AKS [10]), we propose a plug-and-play module that dynamically selects frames to maximize both relevance to textual prompts and temporal coverage of video content. By integrating principles from computer science algorithms—such as recursion, divide-and-conquer, and greedy strategies—our approach offers a theoretically grounded and computationally efficient solution.

1.1. Background and Motivation

Long-form videos (e.g., over one hour) pose unique challenges for MLLMs, as their fixed context windows cannot process every frame. Traditional methods like uniform sampling or top relevance sampling fail to balance relevance and coverage effectively: UNI ignores semantic relevance, while TOP may select clustered frames, missing critical events in other segments. This project aims to bridge this gap by designing a hierarchical keyframe selection algorithm that adapts to video content characteristics.

1.2. Approach and Algorithm Design

The AKS module formulates keyframe selection as an optimization problem balancing two objectives:

1. **Relevance**: Frames must align with the user's textual prompt, measured via vision-language models.
2. **Coverage**: Frames must be temporally distributed to avoid redundancy and ensure full video representation.

To solve this, we employ a **recursive divide-and-conquer strategy** inspired by course algorithms like merge sort:

1.2.1. Recursive Judge-and-Split

The video timeline is partitioned into segments. At each level, if a segment contains frames with significantly higher relevance via a threshold, we use a greedy strategy to select top frames locally. Otherwise, the segment is split into subsegments, recursively applying the same logic.

1.2.2. Adaptive Balance

The algorithm dynamically adjusts between relevance (greedy selection) and coverage (divide-and-conquer) using hyper-parameters, mirroring dynamic programming's state transition logic. This design directly integrates concepts from the course curriculum:

- **Recursion**: The hierarchical splitting of video segments mirrors recursive algorithm structures (e.g., quicksort).
- **Greedy Algorithms**: The top-k frame selection within segments embodies greedy optimization, prioritizing immediate relevance.
- **Divide-and-Conquer**: The temporal partitioning strategy decomposes the problem into smaller subproblems, similar to merge sort or binary search.
- **Complexity Analysis**: The algorithm's time complexity is optimized using partial sorting, reducing per-segment cost from $\mathcal{O}(n \log n)$ to $\mathcal{O}(n)$, aligning with course teachings on algorithm efficiency.

2. Related Work

The rapid growth of user-generated videos has made automated content moderation increasingly important. AI systems are now essential for identifying and filtering inappropriate or harmful content efficiently. A key part of these systems is selecting frames from videos that best represent their content, allowing for effective moderation without processing every single frame.

2.1. AI-Based Video Content Moderation

Traditional video moderation methods often rely on simple classification models trained on fixed categories of unsafe content. These models can be limited in scope and may not provide detailed explanations for their decisions. To address these limitations, researchers have explored more advanced approaches.

Deodhar et al. [2] proposed a human-machine collaboration framework that combines machine learning predictions with human annotations. This approach creates a feedback loop that enhances model performance over time. Gomez et al. [3] highlighted challenges in content moderation, emphasizing the need for transparency and consistency in AI-driven decisions.

A significant advancement in this field is SafeWatch, introduced by Chen et al. [1]. SafeWatch is an efficient video guardrail model that follows customized safety policies and provides detailed explanations for its decisions. Unlike traditional models, SafeWatch processes safety policies in parallel, reducing bias and improving efficiency. It also incorporates a policy-aware visual token pruning algorithm that selects the most relevant parts of a video for analysis, discarding irrelevant information. This approach allows SafeWatch to handle a wide range of unsafe content categories effectively.

2.2. Keyframe Selection in Video Moderation

Selecting the right frames from a video is crucial for efficient content moderation. Analyzing every frame is computationally expensive, so identifying keyframes that represent the video's content is essential.

Tang et al. [10] introduced an Adaptive Keyframe Sampling (AKS) method that formulates keyframe selection as an optimization problem. This method balances the relevance of frames to the moderation task and the overall coverage of the video content.

Patil et al. [8] developed a model combining Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) for video content classification. Their approach includes a novel keyframe extraction method that reduces processing time without sacrificing performance.

Roberts et al. [9] proposed an optimal and interactive keyframe selection method for motion capture. While originally designed for motion sequences, this method can be adapted for content moderation purposes, focusing on frames that best represent the content.

In summary, advancements in AI-based video content moderation and keyframe selection methods have significantly improved the efficiency and accuracy of content analysis. Models like SafeWatch demonstrate the potential of combining policy-aware processing with adaptive frame selection to enhance moderation systems.

3. Methodology

Our approach, termed **AKS**, introduces a plug-and-play keyframe selection module designed to enhance the performance of Multimodal Large Language Models (MLLMs) on long video understanding tasks. This module operates by intelligently selecting a subset of frames that maximizes informational value for the MLLM, given its limited context capacity. The methodology is rooted in optimizing for two crucial aspects: the relevance of keyframes to a given textual prompt and the comprehensive coverage of these keyframes across the video's duration.

3.1. Preliminaries and Problem Formulation

Let a video be represented as $\mathbf{V} \in \mathbb{R}^{T \times W \times H \times C}$, where T is the total number of frames, and W, H, C are the width, height, and number of channels per frame, respectively. Each frame $\mathbf{V}_t$ (for $t \in \{1, 2, \ldots, T\}$) can be processed by a pre-trained visual encoder (e.g., CLIP ViT-L) to extract a set of visual tokens, denoted as $\mathbf{F}_t$. The textual input, or prompt, is denoted as $\mathbf{Q}$.

Standard video-based MLLMs process video content by feeding visual tokens derived from video frames, along with the text prompt, into a large language model. The typical input structure is [**User:** ⟨video-tokens⟩ ⟨text-instruction⟩ **Assistant:**]. We denote the MLLM's understanding capability as a function $G(\{\mathbf{F}_t\})$, focusing on the visual context. Due to the inherent limitations in the context window size of MLLMs, processing all frames from a long video (T can be very large) is infeasible. Thus, a keyframe selection strategy is imperative. The core problem is to design a selection function, $\mathrm{KS}_M(\mathbf{Q}, \mathbf{F})$, that outputs an index set $\mathcal{I} \subseteq \{1, 2, \ldots, T\}$ such that $|\mathcal{I}| = M$, where M is the maximum number of frames the MLLM can process (here $\mathbf{F}$ can be considered the full set of features from all candidate frames). The tokens extracted from these M keyframes, $\{\mathbf{F}_t \mid t \in \mathrm{KS}_M(\mathbf{Q}, \mathbf{F})\}$, are then fed to the MLLM. The efficacy of the MLLM heavily relies on the quality of these selected keyframes.

3.2. Principles of Keyframe Selection

The primary goal of the keyframe selection function $\mathrm{KS}_M(\mathbf{Q}, \mathbf{F})$ is to maximize the useful information content within the selected M frames. This can be abstractly formulated as:

$$\mathrm{KS}_M(\mathbf{Q}, \mathbf{F}) = \arg\max_{|\mathcal{I}|=M} G'(\{\mathbf{F}_t \mid t \in \mathcal{I}\}) \tag{1}$$

where $G'(\cdot)$ represents an idealized measure of the MLLM's confidence or the quality of its output based on the provided keyframes. However, this formulation is mathematically intractable due to the combinatorial search space for $\mathcal{I}$ and the difficulty in defining and optimizing $G'(\cdot)$ directly, especially since ground-truth supervision for optimal keyframes is typically unavailable.

To address this, we propose a heuristic approach based on two intuitive principles:

1. **Relevance**: The selected keyframes should be highly relevant to the input prompt $\mathbf{Q}$. The visual content of these frames should directly contribute to answering the query or performing the instructed task.
2. **Coverage**: The set of selected keyframes should offer sufficient coverage of the important events and information spread across the entire video. This principle aims to prevent redundancy (e.g., selecting multiple near-identical adjacent frames) and ensure that crucial information from different temporal segments is not missed.

Based on these principles, we reformulate the optimization problem as:

$$\mathrm{KS}_M(\mathbf{Q}, \mathbf{F}) = \arg\max_{|\mathcal{I}|=M} \sum_{t \in \mathcal{I}} s(\mathbf{Q}, \mathbf{F}_t) + \lambda \cdot c(\mathcal{I}) \tag{2}$$

Here, $s(\mathbf{Q}, \mathbf{F}_t)$ quantifies the relevance between the prompt $\mathbf{Q}$ and the visual tokens $\mathbf{F}_t$ of the t-th frame. The term $c(\mathcal{I})$ measures the temporal coverage provided by the set of selected keyframe indices $\mathcal{I}$. λ is a hyper-parameter balancing the trade-off between relevance and coverage.

Computing Relevance $s(\mathbf{Q}, \mathbf{F}_t)$: The relevance score $s(\mathbf{Q}, \mathbf{F}_t)$ is computed using an auxiliary, computationally efficient vision-language (VL) model, such as CLIP or BLIP ITM. This model takes the prompt $\mathbf{Q}$ and a candidate frame $\mathbf{V}_t$ (or its features $\mathbf{F}_t$) as input and outputs a similarity score indicating how well the frame's content aligns with the prompt.
Estimating Coverage $c(\mathcal{I})$: Coverage is related to the homogeneity of the selected keyframes' distribution along the video's timeline. We adapt concepts similar to Ripley's K-function. The video's timeline $[0, T)$ is recursively partitioned into bins. For instance, at the first level, it's divided into two bins: $[0, T/2)$ and $[T/2, T)$. If m_1 and m_2 keyframes fall into these respective bins, a penalty (e.g., related to $|m_1 - m_2|$) is implicitly incorporated by ensuring a balanced distribution. This recursive partitioning continues for L levels, where $L \leqslant \lceil \log_2 M \rceil$ is a hyper-parameter. The goal of maximizing $c(\mathcal{I})$ is to encourage the selection of frames that are well-dispersed across these temporal bins, thereby covering different segments of the video.

3.3. Adaptive Keyframe Sampling (AKS)

Solving Equation 2 directly remains challenging due to the complex nature of $c(\mathcal{I})$. AKS provides an adaptive algorithm to approximate the optimal keyframe set.

We first consider simpler baseline strategies derived from specific settings of λ:

- **Uniform Sampling (UNI):** This strategy, often used as a default baseline (e.g., in LLaVA-Video), selects frames at regular intervals across the video, effectively ignoring both $s(\mathbf{Q}, \mathbf{F}_t)$ and $c(\mathcal{I})$ beyond a basic even spread. This corresponds to a degenerate case where $s(\mathbf{Q}, \mathbf{F}_t)$ is constant and coverage is maximized by even spacing.
- **Top Sampling (TOP):** When $\lambda = 0$, coverage is disregarded. The algorithm simply selects the M frames that have the highest relevance scores $s(\mathbf{Q}, \mathbf{F}_t)$. While this maximizes relevance, it may lead to selecting temporally clustered frames, potentially missing crucial information elsewhere in the video.
- **Binned Sampling (BIN):** When $\lambda \to +\infty$, coverage is strictly prioritized. The video is divided into a number of bins (e.g., M bins, or fewer bins with multiple selections per bin), and keyframes are selected based on the highest relevance scores within each bin. This ensures temporal spread but might select frames with globally lower relevance if a bin contains no highly relevant frames.

Our proposed **Adaptive Sampling (ADA)** algorithm offers a more nuanced approach by dynamically balancing relevance and coverage. It employs a hierarchical, recursive optimization strategy:

1. **Initialization**: The algorithm starts with the entire video (or a list of candidate frames sampled at a certain frequency, e.g., 1 fps) as a single segment.

2. **Recursive Judge-and-Split**: At each level of recursion, for a given temporal segment (bin) and a quota of keyframes to select from it:
   - Calculate relevance scores $s(\mathbf{Q}, \mathbf{F}_t)$ for all candidate frames within the current segment.
   - Compute the average score of all frames in the segment (s_{all}) and the average score of the top-k frames (where k is the current quota for this segment, or a proxy for it) (s_{top}).
   - **Decision**: If the difference $s_{\text{top}} - s_{\text{all}}$ exceeds a predefined threshold s_{thr}, or if only one keyframe needs to be selected for this segment, it implies that some frames are significantly more relevant than others. In this case, the algorithm prioritizes relevance and directly selects the top-k highest-scoring frames from this segment (similar to **TOP** within the segment).
   - **Splitting**: Otherwise (if $s_{\text{top}} - s_{\text{all}} \leq s_{\text{thr}}$), it suggests that relevance is more uniformly distributed, or that a specific focus is not clearly identifiable. To ensure coverage, the current segment is split into two sub-segments (e.g., at its temporal midpoint). The keyframe quota is divided (typically evenly) between these sub-segments. The algorithm then recursively calls itself on these sub-segments.
3. **Termination**: The recursion stops when the desired number of keyframes M has been allocated, or a maximum recursion depth L is reached. If a segment is not split further due to reaching max depth, frames are selected based on relevance within that final segment.

The hyper-parameter s_{thr} in the ADA algorithm implicitly controls the trade-off, analogous to λ in Equation 2. A higher s_{thr} makes the algorithm lean more towards splitting (prioritizing coverage, like **BIN**), while a lower s_{thr} makes it lean towards direct top-k selection (prioritizing relevance, like **TOP**). This adaptive mechanism allows AKS to dynamically adjust its selection strategy based on the characteristics of the video content and its relevance to the prompt, leading to a robust selection of informative keyframes. Figure 3 in the original paper (not included here) provides a visual illustration of this adaptive sampling process.

4. Experiments And Improvements

Our method and the detailed improvements can be accessed through the following link: `https://github.com/daxuanzi515/MixedVideoAKS`.

4.1. Experimental Setup and Implementation

Benchmarks. We employ two benchmark datasets for long video understanding and mixed video synthesis, respectively: LongVideoBench [11] and XD-Violence [5]. The former contains videos that can exceed one hour in duration, making the quality of keyframe selection critical for effective visual understanding. The latter comprises two types of short violent video clips—*Fighting* and *Shooting*—each lasting approximately 1 to 1.5 minutes, and serves as the basis for constructing malignant segments in our synthesized mixed videos. In addition, we aim to evaluate the robustness of our method in identifying keyframes within concealed malignant content—referred to as **mixed videos**. To this end, we develop an automated pipeline for mixed video synthesis, which also serves as an additional dataset in our experiments.

Metrics. The datasets include question-answer (QA) pairs, where a single question may correspond to multiple segments of a video, each with several candidate answers. The evaluation baseline is a MLLM that processes selected keyframes (i.e., sampled video frames) along with the corresponding QAs. The model is expected to output responses that may include one or more candidate choices. We assess model performance by calculating the coverage of correct answers, as well as measuring inference time, to evaluate the quality of selected keyframes and the overall efficiency improvement introduced by our method.

Implementation Details. Given computational constraints, we utilize MiniCPM-V 2.6, the most recent and capable model in the MiniCPM-V series—which offers enhanced support for multi-image and video understanding. This model serves both as our evaluation baseline and as the core component of our automated video processing pipeline. Our experiments are conducted on a machine running Ubuntu 20.04, equipped with an NVIDIA RTX 3090Ti GPU (24GB VRAM), 2TB of storage, and Python 3.9 managed via Anaconda. For inference acceleration, we adopt the `vLLM` framework in conjunction with the AWQ quantization technique. Specifically, we utilize the MiniCPM-V 2.6 model [7], which supports multi-image and video understanding, and its AWQ INT4 quantized variant [4] for efficient deployment.

4.2. Dataset Benchmark

To better evaluate our algorithm's capability under realistic and complex scenarios, we propose a synthetic benchmark dataset tailored for violent content detection in long videos. Our dataset construction is aligned with the structure of the feature extraction pipeline introduced in prior work [12], and enhanced with a tailored video synthesis process to simulate real-world moderation tasks.

Specifically, we start by collecting pure violent video content from the XD-Violence dataset [5] and designing guided prompts. We then interact with GPT-4o [6] to generate QA pairs templates that focus on the violent content in these videos and then enhance them with random permutation. These QA annotations are created solely based on the original violent videos, without introducing benign clips at this stage. Then, we employ a controlled synthesis strategy to create long-form mixed videos. Each synthesized video

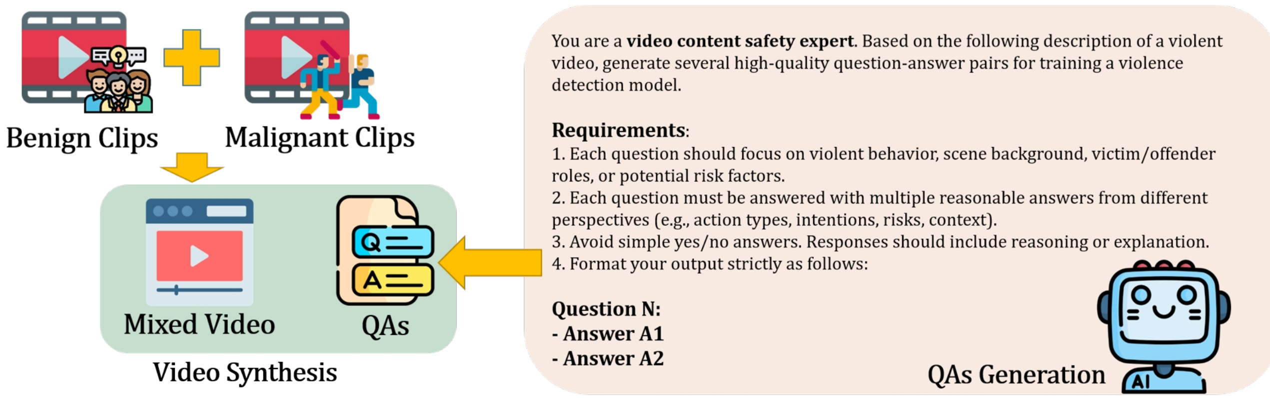

Figure 1. WorkFlow of Mixed Video Synthesis Pipeline

consists of three parts: a benign segment, a malignant segment, and another benign segment, denoted as B_1–M–B_2. This design reflects real-world moderation scenarios where harmful content is often embedded between segments of safe content. We vary the position and ratio of each segment to increase diversity and realism.

We define two mixing strategies for video synthesis:

- **Class 1**: The malignant segment M is placed near the **50%** position of the final video timeline.
- **Class 2**: The malignant segment M is placed near the **75%** position (i.e., centered) in the video.

In both classes, the surrounding benign segments B_1 and B_2 are sampled and trimmed such that the total video length remains approximately 6 minutes, and the inserted M spans about 0.5 or 1 minute. During our analysis of content distribution patterns in real-world datasets, we observed that violent or inappropriate content frequently appears in the second half or final quarter of long videos, motivating our design of Class 1 and Class 2.

Following this pipeline, we construct approximately 480 QA pairs and 120 mixed videos, resulting in a dataset of roughly 7 GB. Each video is paired with the generated QA annotations to support temporal and semantic evaluation tasks. An overview of our data synthesis and QA generation pipeline is illustrated in Figure 1.

4.3. Improvements

To improve the robustness and interpretability of keyframe selection in long video understanding tasks, we propose a revised hierarchical top-k segmentation algorithm. The original implementation recursively splits the score sequence based on heuristic thresholds. However, it lacked error handling and produced hard-to-debug logic branches. Our improved algorithm provides a more modular structure with depth-aware splitting and robust boundary checking.

4.3.1. Optimized Algorithm Description

As Algorithm 1 show, let $S = \{s_1, s_2, \ldots, s_n\}$ be the normalized frame-level scores of a video, and $F = \{f_1, f_2, \ldots, f_n\}$ the corresponding frame identifiers. The objective is to select K representative frames by recursively segmenting S and selecting local top-k values within each segment based on a significance threshold.

Algorithm 1: Hierarchical Keyframe Selection

Input: Score list S, frame list F, target number K, thresholds t_1, t_2, max depth D
Output: Combined selected frames
Initialize queue $Q \leftarrow [(S, F, 0)]$, selected list $L \leftarrow []$;
while *Q is not empty* **do**
  Pop (s, f, d) from Q;
  Compute $\mu \leftarrow \text{mean}(s)$, $\sigma \leftarrow \text{std}(s)$;
  Let $k \leftarrow \min(K, |s|)$, $T_k \leftarrow \text{top-}k(s)$;
  if *$mean(T_k) - \mu > t_1$ **and** $\sigma > t_2$* **then**
    Append (s, f, d) to L;
  else if *$d < D$ **and** $|s| > 1$* **then**
    Split into $(s_1, f_1), (s_2, f_2)$ at midpoint;
    Enqueue $(s_1, f_1, d+1)$ and $(s_2, f_2, d+1)$ into Q;
  else
    Append (s, f, d) to L;
From each $(s, f, d) \in L$, select top-$\min(K/2^d, |s|)$ frames;
return Combined selected frames;

4.3.2. Improvements Over the Original

Compared to the original version, our revised implementation introduces:

- Safe handling for edge cases where $K > |s|$, preventing runtime errors in top-k selection.
- Depth-controlled recursive splitting to maintain segment granularity.
- Transparent logging of mean/std deviation and top-k difference to enable better diagnostics.

- Deterministic frame count allocation per segment based on recursion depth, enforcing global quota K.

4.3.3. Time Complexity Analysis

Let n be the number of input frames. The complexity comparison is as follows:

- **Original:** In worst case, recursively splits the score list up to d levels with repeated sorting, yielding $\mathcal{O}(d \cdot n \log n)$ due to repeated use of heapq.nlargest.
- **Optimized:** Uses argpartition for partial top-k retrieval, with cost $\mathcal{O}(n)$ per segment and $\mathcal{O}(n \log d)$ overall in the balanced case.

Therefore, the revised algorithm reduces per-segment selection cost from $\mathcal{O}(k \log n)$ to $\mathcal{O}(n)$, improving runtime performance and scalability. In the worst case, the algorithm requires $\mathcal{O}(n \log k)$ operations due to repeated top-k selection. With early stopping and approximate selection (e.g., QuickSelect), the average-case complexity reduces to $\mathcal{O}(n)$.

4.3.4. Correctness and Rationality

The revised algorithm preserves the core idea of selecting statistically significant segments with local saliency. By measuring deviation from global mean and standard deviation, it adapts the granularity of segments based on content variance. The total number of selected frames is constrained by the user-defined maximum K, maintaining consistent output length. The empirical comparison further confirms that, while the selected frame counts remain equal, the actual selected content differs—reflecting more refined decision boundaries due to structured depth control.

4.4. Evaluation

An example of our evaluation for the two types of violent actions is illustrated in Figure 2. The testing model selects keyframes from the entire video and leverages the provided question along with multiple candidate choices to complete the task. The model is allowed to produce only one correct answer at a time.

4.4.1. Frames Comparison

To better evaluate the quantity of selected frames, we compare the frame refinement capabilities of SeViLA, AKS, and our method, MixedVideoAKS (MAKS). We use three metrics for comparison: frames count, different entries, and average overlap. These metrics respectively represent the selection scale, selection content, and selection overlap. A lower frames count indicates stronger frame selection, while different selection contents and overlap show that the methods extract distinct frames during the process.

As shown in Table 1, both AKS and MixedVideoAKS can select fewer frames from the original set, thereby reducing redundant content. The overlap between AKS and MixedVideoAKS indicates that these methods select different frames as final keyframes. Although they have a high overlap, MixedVideoAKS can handle more complex inputs and reduce repetitive work.

Table 1. Frames Comparison from Extracted Frames

Comparison Pairs	Frames Count	Different Entries	Average Overlap
AKS VS. SeViLA	7680/41289	120.00	0.1892
MAKS VS. SeViLA	7680/41289	120.00	0.1892
MAKS VS. AKS	7680/7680	**7.25**	**0.9951**

4.4.2. Testing Task Correctness

Our evaluation metrics are as follows:

- **General Correct Rate**. This metric indicates the overall accuracy of the selected frames in the testing task. It includes frames containing keywords such as "Fighting", "Shooting", or "NONE". Since we select frames that contain both benign and malignant contents, it is challenging to completely avoid benign contents. To address this, we consider answers containing irrelevant words (e.g., "UNKNOWN" or incorrect keywords) as entirely wrong. For instance, the answer should not include "Shooting" during a "Fighting" class task.
- **True Positive Rate**. This metric measures the proportion of completely correct answers in the testing task. For example, the model should provide the answer "Fighting" for frames that are labeled as "Fighting".
- **False Positive Rate**. This metric indicates the proportion of answers that are incorrectly identified as "NONE". While the testing model may not find any keywords in the given videos, this is permissible.
- **True Negative Rate**. This metric measures the proportion of answers that are disallowed or marked as "UNKNOWN", which are considered incorrect.

Our evaluation encompasses two distinct mixed video scenarios: "Fighting" and "Shooting", each characterized by two unique mixing proportions. This comprehensive assessment allows us to thoroughly analyze the performance of our proposed method, MixedVideoAKS, against existing solutions. The original algorithm results are provided in parentheses for reference. As illustrated in Table 2, we conducted a detailed comparison between our MixedVideoAKS method and the AKS algorithm to evaluate the accuracy of our approach. The results from four different types of mixed videos clearly demonstrate that MixedVideoAKS can significantly enhance the true positive rate while effectively reducing the false positive rate during the testing phase. This improvement underscores the robustness and reliability of our method in various complex video environments.

To further validate the effectiveness of MixedVideoAKS, we reconstructed a series of comparative continuous videos using the keyframes extracted by our method. When com-

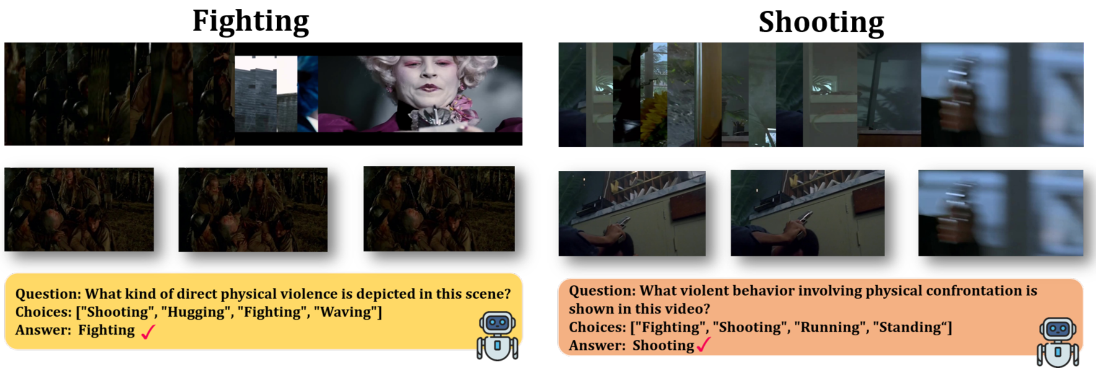

Figure 2. Example of Fighting and Shooting Evaluation

Table 2. Video-based question answering accuracy (%) of different approaches on Mixed Videos.

Method	Task Type	General Correct Rate (%)	True Positive Rate (%)	False Positive Rate (%)	True Negative Rate (%)
MixedVideoAKS	Fighting_Class_1	94.25↑(86.21)	43.68↑(40.23)	50.57↓(45.98)	6.90↓(13.79)
	Fighting_Class_2	93.33↑(85.56)	27.78↑(24.44)	65.56↓(66.67)	6.67↓(8.89)
	Shooting_Class_1	97.78↑(88.89)	46.67↑(44.44)	51.11↓(44.44)	2.22↓(5.56)
	Shooting_Class_2	97.75↑(91.01)	49.44↑(46.07)	48.31↓(44.94)	2.25↓(3.37)

pared with the initial source videos, it was found that over 50% of our keyframes could be matched to the original content. This high matching rate indicates that our key content is well-represented and effectively captured, demonstrating the robustness of our method in identifying and preserving the most important frames.

Overall, our evaluation demonstrates that MixedVideoAKS outperforms the original AKS algorithm in terms of both true positive and false positive rates. This enhanced performance, coupled with the method's ability to handle complex video environments, positions MixedVideoAKS as a valuable tool for advanced video analysis tasks. Future work will focus on further optimizing the algorithm and exploring additional applications in various video processing domains.

5. Conclusion

This paper introduces a novel plug-and-play keyframe selection module, which effectively optimizes keyframe selection for long video understanding tasks by balancing relevance and temporal coverage. Through the integration of advanced algorithmic principles such as recursion, divide-and-conquer, and greedy strategies, our module achieves a substantial reduction in computational complexity. The experiments on LongVideoBench and a self-constructed mixed video dataset show that the proposed approach outperforms traditional sampling methods. Furthermore, a revised hierarchical top-k segmentation algorithm improves the module's robustness and interpretability. By optimizing keyframe selection, our module not only improves the efficiency of video processing but also enhances the overall performance of MLLMs in understanding and moderating long-form videos.

6. Contribution Percent

Our team operated with a collaborative framework, featuring overlapping responsibilities and mutual support across all tasks. The specific responsibilities and cross-functional contributions are outlined as follows:

Chen Xuanxin: Oversaw the Experiments & Improvement section, including dataset construction , algorithm coding, and slide preparation for experimental results. Her technical leadership ensured robust validation of the keyframe selection method.

Zhenzhen Xu: Led the development of the Abstract and Introduction , managed report editing, and coordinated slide design. Contributed to dataset construction and annotation, ensuring high-quality labeling for mixed video benchmarks. Her expertise in technical writing and narrative structuring enhanced the project's communicative clarity.

Wu Hao: Spearheaded the Methodology section, detailing algorithm design, mathematical formulations, and complexity analysis. Participated in dataset construction and annotation, providing domain-specific insights for violent content segmentation.

Qinzhou Zhang: Contributed to the Related Work section, conducting a comprehensive literature review and comparative analysis of state-of-the-art methods like SafeWatch and AKS. Actively supported dataset construction and annotation, focusing on semantic consistency across video segments.

Xiaolong Guo: Managed the Conclusion section and final slide development, summarizing key findings and designing visualizations. Played a critical role in dataset construction and annotation, ensuring accurate labeling of benign/malignant segments in mixed videos.

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